Predicting Coronary Heart Disease from BRFSS Survey Data with Linear and Logistic Models

Hassen Ben Chaabane, Meriem Laataoui, Youssef Ben Lassoued
Machine Learning (CS-433), EPFL, Switzerland

***Abstract*—[Abstract: task, data, methods, nested-CV F1 of the final model, main finding (codebook preprocessing matters most; class weighting and regularization do not help an underfitting linear model).]**

I. Introduction

Cardiovascular disease is the leading cause of death worldwide. We predict self-reported coronary heart disease or myocardial infarction (MICHD) from the 2015 BRFSS telephone survey, a binary task with strong class imbalance, scored with F1 and solved with NumPy only. We contribute (a) a label-free preprocessing built from the BRFSS codebook (Section II); (b) a nested cross-validation protocol in which the threshold is also chosen without seeing the test fold (Section III-A); (c) a comparison of linear and logistic models and of class weighting, and a final model with interactions (Sections III-C–III-D); and (d) ablations showing which choices actually improve F1 (Section IV).

II. Data and preprocessing

[Person 1, data description: training and test sizes, class balance, missing values, feature types. Numbers from the shared EDA: 328,135 training and 109,379 test respondents, 321 features, 8.8% positive, 44.8% of all values missing, 147 features missing for more than half of the respondents, mostly from optional state modules. Optional: EDA figure.]

The features are coded survey answers, so we first convert them into meaningful values with rules from the BRFSS 2015 codebook [1], checked for every feature (Table I). The rules depend only on the feature values, never on the labels.

[Person 1, one or two sentences: median imputation, standardization, bias column; all statistics learned on training rows only. Applied alone, this is the generic baseline of Table III.]

Rule	Example
Don't know, refused → missing	codes 7 and 9
None → 0	codes 88, 888, 555; 62% answer 88 = "no days" of poor physical health
Drop 32 survey features	identifiers, weights
Drop 29 raw answers	WEIGHT2, cleaned in WTKG3
Drop 3 nominal features	activity types, kept in computed minutes; one 99.9% missing

Table I
Codebook rules, applied before any learning.

The final model adds two kinds of binary features: a one-hot encoding of the nominal features and of every feature with 3 to 10 values, and, for each feature with missing values, an indicator of missingness. The indicators are informative because many questions are only asked after a given answer: the age at diabetes diagnosis, for instance, is missing for respondents without diabetes. **[Person 1: own improvement, if any.]**

III. Methods

A. Evaluation protocol

Models are evaluated by nested cross-validation, so that selection and evaluation use disjoint data. The training set is split into five stratified folds, shared by all experiments. Each fold serves once as the test fold; on the four others, the preprocessing statistics are estimated and an inner 3-fold cross-validation selects the hyperparameters and the threshold t that maximize F1, after which the selected model is retrained on them and evaluated on the test fold. We report the mean and standard deviation of F1 over the test folds. Two configurations are compared by their per-fold F1 difference Δ, which removes the variation due to the split, and one is considered better only if $\Delta > 0$ in all five folds. The submitted model results from the same selection on the full training set.

B. Linear models

[MSE gradient descent, SGD (batch size 1), least squares, ridge: losses, grids, iterations, seed; continuous scores thresholded like probabilities.]

C. Logistic models

Logistic regression models the probability of disease as $\sigma(x_i^\top w)$, with $y_i \in \{0, 1\}$, and minimizes the mean negative log-likelihood $L(w) = \frac{1}{N}\sum_i \big[\log(1 + e^{x_i^\top w}) - y_i x_i^\top w\big]$; its regularized version minimizes $L(w) + \lambda\|w\|^2$. Both use full-batch gradient descent with step size γ from $w = 0$ for 300 iterations, after which the loss changes by less than 0.06%. The inner cross-validation selects $\gamma \in \{0.01, 0.03, 0.1, 0.3, 1\}$, $\lambda \in \{10^{-6}, \ldots, 10^{-1}\}$ and $t \in \{0.05, 0.06, \ldots, 0.95\}$.

D. Final model

A linear model cannot capture joint effects, such as smoking being more dangerous for older respondents. The final model therefore adds the products of the K most important features. Regularized logistic regression, which has the highest F1 (Table II), is first fitted to rank the standardized features by $|w_j|$, then refitted with the $K(K-1)/2$ standardized products of the top K. $K \in \{20, 30, \ldots, 80\}$ is chosen by 3-fold cross-validation, and γ, λ and t as before.

IV. Results

The final model is the best method (Table II), with F1 $= 0.441 \pm 0.006$ in nested cross-validation and 0.454 on AIcrowd. Its gain does not come from the hyperparameters: the inner selection gives $\gamma = 0.3$, $\lambda = 10^{-4}$ and, for the final model, $K = 40$, but F1 is flat around these values (Fig. 1) and every outer fold selects $\gamma \in \{0.3, 1\}$ and $\lambda \leq 10^{-3}$. Nor does it come from regularization: plain and regularized logistic regression are tied, $\Delta = 0.0003 \pm 0.0014$, since with training and validation losses of 0.219 and 0.221 the model does not overfit. **[Person 3: linear vs. logistic comparison once Table II is complete.]**

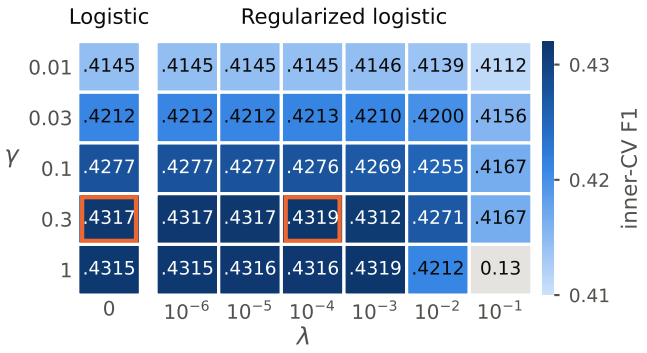


Figure 1. Inner-CV F1 on the full training set; $\lambda = 0$: plain logistic regression. F1 rises with γ until gradient descent converges and does not depend on $\lambda \leq 10^{-3}$. Orange: selected.

Method	F1	Prec.	Rec.	Acc.	Thr.
GD (MSE)	[]				
SGD (MSE)	[]				
Least squares	[]				
Ridge	[]				
Logistic	0.431 ± 0.005	0.359	0.540	0.874	0.20
Reg. logistic	0.432 ± 0.005	0.358	0.543	0.874	0.20
Final model	$\mathbf{0.441 \pm 0.006}$	0.369	0.549	0.877	0.21

Table II
NESTED CROSS-VALIDATION, MEAN OVER THE 5 TEST FOLDS. THR.: MEAN SELECTED THRESHOLD. ALL ROWS BUT THE LAST ONE-HOT ENCODE THE NOMINAL FEATURES ONLY. ALWAYS PREDICTING "NO DISEASE" GIVES ACCURACY 0.912 BUT F1 = 0.

Configuration	F1	Δ
Generic baseline	0.416 ± 0.005	
+ codebook rules	0.429 ± 0.003	$+0.013 \pm 0.004$
+ one-hot encoding	0.433 ± 0.005	$+0.004 \pm 0.003$
+ missing indicators	0.438 ± 0.004	$+0.005 \pm 0.002$
+ 40-feature interactions	0.441 ± 0.006	$+0.003 \pm 0.002$

Table III
FROM THE BASELINE TO THE FINAL MODEL, ONE CHANGE PER ROW. Δ: PER-FOLD GAIN OVER THE ROW ABOVE. AFTER THE BASELINE, $\gamma = 0.3$ AND $\lambda = 10^{-4}$.

The gain therefore comes from the features, which Table III adds one at a time. The codebook rules contribute most, through precision, which rises from 0.337 to 0.361. One-hot encoding and missing indicators then add 0.004 and 0.005 in all folds, and the interactions 0.003 in four folds only, a likely but not certain gain. Other additions, such as squared or binned features, more interactions or model averaging, changed F1 by less than 0.001.

Finally, the class imbalance needs no correction beyond the threshold. Multiplying the loss of the positive samples by 3.2 or 10.3, the ratio of negatives to positives, barely changes F1, 0.431 and 0.430 against 0.432, but raises the selected threshold from 0.21 to 0.48 and 0.75. Weighting raises all predicted probabilities while keeping their order, so a higher threshold gives back the same predictions: as the odds are multiplied by about ω, 0.21 becomes 0.73 for $\omega = 10.3$.

V. Conclusion

The final model, regularized logistic regression with interactions, reaches a nested-CV F1 of 0.441 ± 0.006 and 0.454 on AIcrowd. The codebook rules give the largest gain, the model does not overfit, so regularization does not help, and threshold tuning already handles the imbalance. The labels are self-reported and the survey is observational, so the model captures associations, not causes, and with a precision of 0.37 most predicted positives are false alarms. The small, uncertain gain of the interactions suggests that linear models are close to their limit on these features.

Reproducibility

Code: **[public GitHub URL and commit]**. `run.py` reproduces the submitted predictions. `tune_models.py`, `evaluate_best.py` and `imbalance.py` produce Tables II and III, and `plots.py` draws Fig. 1; the README lists all commands.

References

[1] Centers for Disease Control and Prevention, "Behavioral risk factor surveillance system: LLCP 2015 codebook report," https://www.cdc.gov/brfss/annual_data/2015/pdf/codebook15_llcp.pdf, 2016, accessed October 2026.